

Codex Prompt — Nursing Reviewer Website

Build a Complete Nursing Student Reviewer Website

You are an expert full-stack web developer, UI/UX designer, database architect, and educational-platform developer.

Build a complete, production-quality nursing reviewer website designed specifically for a nursing student.

The website should feel like a modern study platform similar in functionality to popular flashcard/reviewer apps, but it must have its own original branding, UI, layout, and design.

The website should be responsive and work extremely well on phones because the primary user will likely study using a mobile device.

Do not create only a static mockup. Build the actual functional application with frontend, backend, database, authentication, reviewer system, quiz system, progress tracking, and an admin/content-management system.

1. Technology Stack

Use:

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui or another clean component system
- Lucide icons
- Next.js server actions/API routes
- Supabase PostgreSQL
- Prisma if appropriate, otherwise Supabase client directly
- Supabase Auth
- Supabase Storage
- OpenAI API
- Vercel-compatible deployment

Use environment variables for all secrets. Create a .env.example file.

2. Overall Product

Create a nursing study platform called "NurseMate".

Major areas:

1. Dashboard

2. Subjects

3. Flashcards
4. Quiz Center
5. Smart Review
6. Study Materials
7. Progress
8. Profile
9. Admin Dashboard

The application should feel like a combination of nursing reviewer, flashcard platform, quiz platform, personal study tracker, and AI study assistant.

3. Visual Design

Create a clean, modern nursing-student aesthetic:

- White/light backgrounds
- Soft pink
- Soft purple
- Soft blue
- Subtle gradients
- Rounded cards
- Clean typography
- Minimal shadows
- Medical/nursing-inspired icons

Do not make it childish. Make it cute enough to feel personal, professional enough for nursing school, and clean enough for long study sessions.

4. Landing Page

Create a landing page with:

"Nursing made easier. Study smarter. Become the nurse you want to be."

Subtitle:

"A personalized nursing reviewer built to help you learn, practice, and track your progress."

Buttons:

- Start Studying
- Explore Reviewers

Feature cards:

- Smart Flashcards
- Nursing Quizzes

- Study Materials
- Progress Tracking
- AI Reviewer
- Study Streaks

5. Authentication

Implement:

- Sign up
- Login
- Logout
- Forgot password
- Reset password
- Profile setup

Profile fields:

- Full name
- Display name
- Email
- Year level
- School
- Profile picture
- Study goal

Year levels:

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year
- Review/Graduate

Protect authenticated pages and redirect unauthenticated users to login.

6. Dashboard

After login, show:

"Hi, [Name] ☺"

"Ready to study today?"

Show:

- Current streak
- Total study time
- Questions answered
- Average quiz score

- Flashcards mastered
- Overall progress

Today's Goal:

"Complete 20 review questions"

Progress example:

12 / 20

Buttons:

- Continue Studying
- Quick Quiz
- Review Flashcards

7. Nursing Subjects

Create a subject library.

Default subjects:

Fundamentals of Nursing:

- Nursing process
- Vital signs
- Health assessment
- Infection control
- Patient safety
- Documentation
- Communication
- Basic nursing skills

Anatomy and Physiology:

- Cardiovascular system
- Respiratory system
- Nervous system
- Digestive system
- Renal system
- Endocrine system
- Musculoskeletal system
- Reproductive system

Pharmacology:

- Drug classifications
- Dosage calculations
- Routes of administration
- Medication safety
- Side effects

- Contraindications
- Drug interactions
- High-alert medications

Medical-Surgical Nursing:

- Cardiovascular
- Respiratory
- Neurological
- Gastrointestinal
- Renal
- Endocrine
- Musculoskeletal
- Oncology

Maternal and Child Nursing:

- Pregnancy
- Prenatal care
- Labor and delivery
- Postpartum
- Newborn care
- Pediatric conditions

Community Health Nursing:

- Epidemiology
- Health promotion
- Disease prevention
- Community assessment
- Public health programs

Psychiatric Nursing:

- Mental health assessment
- Therapeutic communication
- Anxiety
- Depression
- Psychosis
- Substance use
- Crisis intervention

Allow administrators to add/edit/delete subjects and topics.

8. Flashcard System

Build a fully functional flashcard system.

Each flashcard:

- Question/front
- Answer/back
- Explanation
- Subject
- Topic
- Difficulty
- Tags
- Source/reference
- Created date

Interaction:

- Again
- Hard
- Good
- Easy

Track performance and implement spaced-repetition scheduling.

9. Smart Review

Determine what the student needs to review based on:

- Previous incorrect answers
- Difficulty
- Last review date
- Number of successful answers
- Mastery percentage

Prioritize weak topics.

10. Quiz System

Question types:

1. Multiple Choice
2. True/False
3. Select All That Apply
4. Identification
5. Prioritization
6. Patient Scenario

Each question supports:

- Question

- Options
- Correct answer
- Explanation/rationale
- Difficulty
- Subject
- Topic
- Tags
- Source/reference

After answering, show Correct/Incorrect and the rationale. Do not reveal the answer before submission.

11. Nursing Scenario Questions

Make patient scenarios a major feature.

Include realistic patient information, vital signs, symptoms, lab results, medication information, and nursing assessment.

Create a visually distinct patient scenario card.

12. NCLEX-Style Mode

Create a dedicated "NCLEX Practice" section with:

- Random questions
- Timed quizzes
- Question counter
- Progress bar
- Score
- Rationales
- Review incorrect answers

Do not claim questions are official NCLEX questions. Label them "NCLEX-style practice question."

13. Quiz Configuration

Allow:

- Subject
- Topic
- Number of questions: 5, 10, 20, 50
- Difficulty: Easy, Medium, Hard, Mixed
- Question type
- Randomize Questions
- Randomize Answers

14. Quiz Results

Show:

- Score
- Percentage
- Correct answers
- Incorrect answers
- Accuracy
- Time spent
- Weak topics
- Strong topics

Buttons:

- Review Mistakes
- Try Again
- New Quiz
- Back to Dashboard

Save every quiz attempt.

15. Progress Tracking

Create a detailed progress page.

Show overall mastery and subject mastery.

Charts:

- Quiz scores over time
- Questions answered
- Study time
- Subject mastery
- Weekly activity

16. Study Streak

Track:

- Current streak
- Longest streak
- Last study date

A study day is earned by completing a meaningful study action such as a quiz, flashcard review, or minimum study session.

Prevent duplicate streak increments on the same day.

17. Study Timer

Create:

- Start
- Pause
- Resume
- Reset

Default:

25 minutes study

5 minutes break

Allow custom durations and save completed study sessions.

18. AI Study Assistant

Create "NurseMate AI".

It can answer questions such as:

- What is hypokalemia?
- Explain heart failure in simple terms.
- What are the signs of hypoglycemia?
- Give me 10 questions about antibiotics.

Provide:

- Clear explanation
- Nursing relevance
- Important signs/symptoms
- Nursing considerations
- Memory tips

Include a disclaimer that AI-generated information should be verified with nursing textbooks, instructors, and trusted clinical references, and that the tool is for education rather than professional medical advice.

19. AI Flashcard Generator

Allow text entry or study-material upload.

Generate:

- Summary
- Flashcards
- Quiz questions
- Key terms

Allow preview before saving. Do not automatically save AI-generated content without confirmation.

20. PDF Upload

Create Study Materials:

- PDF upload
- View materials
- Delete materials
- Categorize by subject/topic
- Display filename, subject, and upload date

Validate file type and size. Store files securely.

21. AI Generation Safety

- Do not invent citations.
- Do not fabricate medical facts.
- Clearly identify AI-generated material.
- Encourage verification against trusted nursing resources.
- Never give personalized diagnosis or treatment recommendations.
- Never claim official NCLEX endorsement.

22. Admin Dashboard

Admin-only:

- Add/edit/delete subjects
- Add/edit/delete topics
- Add/edit/delete flashcards
- Add/edit/delete quiz questions
- Upload study materials
- View users
- View statistics

Roles:

student

admin

Do not rely only on frontend authorization.

23. Content Management

Forms for:

Subject:

- Name
- Description
- Icon
- Color
- Order

Topic:

- Name
- Subject
- Description

Flashcard:

- Front
- Back
- Explanation
- Subject
- Topic
- Difficulty
- Tags
- Source

Question:

- Question
- Type
- Options
- Correct answer
- Explanation
- Subject
- Topic
- Difficulty
- Source

24. Database

Suggested tables:

- profiles
- subjects
- topics
- flashcards
- flashcard_progress
- questions
- question_options
- quiz_attempts
- quiz_answers
- study_sessions
- study_materials
- study_streaks
- ai_generations
- user_settings

Use UUIDs where appropriate, foreign keys, timestamps, indexes, user ownership, and role permissions.

Create complete Supabase SQL schema/migrations.

25. Security

Implement:

- Authentication
- Authorization
- Supabase Row Level Security
- Secure API routes
- Server-side API key handling
- Input validation
- File validation
- Rate limiting where appropriate

Never expose private keys in client code.

26. Mobile Design

Mobile-first.

Support:

- Android
- iPhone
- Tablet
- Desktop

Mobile bottom navigation:

Home

Review

Quiz

Progress

Profile

Desktop sidebar.

27. Navigation

Desktop:

Dashboard

Subjects

Flashcards

Quizzes

Smart Review

Materials

Progress

AI Assistant

Profile

Admin

28. Personalization

Dashboard message:

"Keep going, future nurse! ☑☑"

Allow customizable encouragement such as:

"Study hard, love. I'm proud of you. ♥"

Keep romantic content subtle.

29. Dark Mode

Support:

- Light
- Dark
- System

30. Search

Search subjects, topics, flashcards, questions, and study materials.

Filters:

Subject

Topic

Difficulty

Content type

31. Favorites

Allow favorite flashcards, questions, topics, and study materials.

32. Mistake Bank

Save incorrectly answered questions.

Show:

"Questions You Need to Review"

Allow review, removal, and retry. Sort by most frequently missed.

33. Weak Topic Detection

Calculate topic mastery and recommend weak areas.

34. Gamification

Award XP for:

- Flashcards
- Quizzes
- Streaks
- Studying
- Improvement

Levels:

- Nursing Student
- Study Buddy
- Future Nurse
- Clinical Ready
- Nursing Pro

Keep optional.

35. Achievements

Examples:

- First Quiz
- 7 Day Streak
- 100 Flashcards
- 500 Questions
- Pharmacology Master
- Cardiac Expert
- 10 Hours Studied

36. Notifications

Support:

- Study reminders
- Review due
- Streak reminder
- Quiz results
- New study materials

Only send notifications with permission.

37. Accessibility

Use:

- Keyboard navigation
- Proper labels
- Accessible buttons
- Good contrast
- Screen-reader-friendly structure
- Focus states

38. Loading and Error States

Every major page needs loading, empty, and error states.

39. Sample Content

Seed at least:

- 5 subjects
- 3-5 topics per subject
- 5+ flashcards per topic
- 5+ quiz questions per topic

Use original educational sample content. Do not copy copyrighted textbook questions.

40. Medical Accuracy

Do not invent medical facts.

Use established nursing concepts. Where sources are included, use a source field. Do not fabricate page numbers or citations.

41. UX

Use subtle animations:

- Card hover
- Button transitions
- Flashcard flip
- Progress animation
- Quiz feedback
- Achievement unlock

Keep the application fast.

42. Project Structure

Use a clean structure with:

- app/
- components/
- lib/
- hooks/
- types/
- services/
- supabase/
- public/
- styles/

Separate UI, database functions, authentication, AI services, quiz logic, flashcard logic, validation, and types.

43. Reusable Components

Create:

Button

Card

Modal

Dialog

Input

Select

ProgressBar

Flashcard

QuizQuestion

QuizOption

SubjectCard

TopicCard

StatCard

StudyTimer

Navbar

Sidebar

MobileNav

AchievementCard

EmptyState

LoadingState

44. Error Handling

Use user-friendly errors. Never expose database errors, API keys, stack traces, or internal server information.

45. Environment Variables

Create .env.example with:

```
NEXT_PUBLIC_SUPABASE_URL=  
NEXT_PUBLIC_SUPABASE_ANON_KEY=  
SUPABASE_SERVICE_ROLE_KEY=  
OPENAI_API_KEY=
```

Never commit real secrets.

46. README

Create a complete README covering:

- Overview
- Features
- Tech stack
- Installation
- Environment variables
- Supabase setup
- Database migration
- Local development
- AI configuration
- Admin setup
- Vercel deployment

Include exact commands.

47. Testing

Test:

- Authentication
- Flashcard scheduling
- Quiz scoring
- Streak calculation
- Progress calculation
- Question randomization
- Authorization

Fix TypeScript, lint, and production build errors.

48. Seed Data

Create a database seed script with:

- Sample admin
- Sample student
- Nursing subjects

- Topics
- Flashcards
- Quiz questions

Do not hard-code production passwords.

49. Development Workflow

Follow this order:

1. Initialize project
2. Configure TypeScript
3. Configure Tailwind/UI
4. Configure Supabase
5. Create database schema
6. Configure authentication
7. Create application layout
8. Create dashboard
9. Create subjects/topics
10. Create flashcards
11. Create spaced repetition
12. Create quiz engine
13. Create quiz results
14. Create progress tracking
15. Create streak system
16. Create study timer
17. Create study materials
18. Create PDF upload
19. Create AI assistant
20. Create AI flashcard/quiz generator
21. Create admin dashboard
22. Add search/favorites/mistake bank
23. Add achievements/gamification

24. Add responsive mobile UI

25. Add accessibility

26. Add error/loading states

27. Add tests

28. Seed sample data

29. Run lint

30. Run tests

31. Run production build

32. Fix all errors

33. Document setup

50. Important Codex Instructions

Work incrementally.

Do not generate a fake interface and claim the application is complete.

After each major feature:

1. Check code

2. Check TypeScript

3. Check imports

4. Check database queries

5. Check authentication

6. Check mobile responsiveness

7. Fix errors before continuing

Choose simple, maintainable solutions. Keep architecture scalable. Avoid unnecessary dependencies and giant files.

51. Final Quality Standard

The finished website should feel like a real nursing study product, not a school-project mockup.

It should be:

- Fast

- Clean

- Responsive

- Professional
- Cute
- Personalized
- Accessible
- Secure
- Maintainable

The student should be able to:

- Register
- Log in
- Choose nursing subjects
- Study flashcards
- Take quizzes
- Receive rationales
- Review mistakes
- Track progress
- Maintain a streak
- Upload study materials
- Generate AI reviewers
- Use the AI study assistant
- See achievements

Admins should be able to manage educational content.

52. Final Personalized Touch

Dashboard section:

"💡 A little message for you"

Example:

"Keep going, future nurse. I know you're tired sometimes, but I believe in you. One topic at a time. One quiz at a time. You've got this. 💡❤️"

Make the message editable from the admin dashboard while keeping it visually subtle.

53. Start Building

Inspect the current project directory.

If empty, initialize the application.

Implement step-by-step.

Do not ask me to manually create files unless absolutely necessary. Create the required files yourself.

Run available checks/build commands and fix errors.

At the end, provide a concise summary of:

- What was built
- Database setup required
- Environment variables required
- How to run locally
- How to create the admin account
- Remaining configuration

The goal is a fully functional nursing reviewer website, not merely a prototype.